

LAB 1 – USE-CASE FLOW

Smart Parking Space Rental & License Validator

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1. User Registration / Login

1. User opens the Smart Parking application.
2. User enters login or registration details.
3. System validates the details.
4. System allows the user to access the application.

2. Search Parking Space

1. User enters the required location.
2. System searches for available parking spaces.
3. System displays available parking spaces with details such as location, price, and availability.
4. User selects a suitable parking space.

3. Rent / Book Parking Space

1. User selects the parking space.
2. User chooses the required date and time.
3. System checks parking availability.
4. System confirms the booking.
5. User completes the payment.
6. System generates a booking confirmation.

4. License Validation

1. User provides the vehicle license/registration number.
2. System receives the license information.
3. System validates the license details.
4. System checks whether the vehicle is authorized.
5. System displays the validation result.

5. Parking Entry

1. Vehicle arrives at the selected parking location.
2. System identifies the vehicle/license number.
3. System validates the booking and license.
4. If valid, the system allows entry.
5. If invalid, the system denies entry and displays an appropriate message.

6. Parking Exit

1. User requests to leave the parking space.
2. System verifies the booking and parking duration.
3. System calculates the parking charges if required.
4. User completes any pending payment.
5. System records the parking session.
6. System allows the vehicle to exit.

7. Manage Parking Space

1. Parking owner/admin logs into the system.
2. Admin adds or updates parking space details.
3. Admin sets availability and rental charges.
4. System stores the updated information.
5. Users can view the updated parking information.

8. Booking Cancellation

1. User selects an existing booking.
2. User requests cancellation.
3. System checks the cancellation conditions.
4. System cancels the booking.
5. System updates the parking availability.
6. Refund is processed if applicable.

- **Main Flow**

User → Search Parking → Select Parking Space → Check Availability → Book Parking → Payment
→ License Validation → Parking Entry → Parking Session → Parking Exit → Payment/Confirmation

- **Alternative Flow**

Invalid License → License Validation Failed → Entry Denied

Parking Unavailable → Booking Failed → User Searches for Another Parking Space

Payment Failed → Booking Not Confirmed → User Retries Payment